Math 3IA3 - Midterm

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Problem 1

Let S be a subset of $\mathbb{R}$ which is bounded below. Prove that a lower bound w of S is the infimum of S if and only if for every $\epsilon > 0$, there exists some $t \in S$ such that $t < w + \epsilon$.

Solution:

Proof.

(I) Prove the forward direction.

If $w = \inf S$, then $w \leq s, \forall s \in S$ and for all $l \in \mathbb{R}$ such that l lowerbounds S, $l \leq w$ by definition of infimum.

Now, let $\epsilon > 0$ be arbitrary, and assume, for the purposes of deriving a contradiction, that there exists no element $s_0 \in S$ such that $s_0 < w + \epsilon$. This therefore implies that $w + \epsilon$ must be a lower bound of S since $\forall s \in S, w + \epsilon \leq s$. However, clearly $w < w + \epsilon$, which implies that there exists a lower bound of S which is larger than w. This contradicts our prior assumptions that w must be the greatest lower bound of S, and thus it must be the case that for arbitrary $\epsilon > 0$, $\exists s \in S$ such that $s < w + \epsilon$.

(II) Prove the reverse direction.

If w is a lower bound of S with the property that $\forall \epsilon > 0$, $\exists s_0 \in S$ such that $s_0 < w + \epsilon$, this implies that $s_0 - \epsilon < w$.

Let l be an arbitrary lower bound of S. Then we know for l to lower bound S, $l \leq s_0$, however, this implies that $l < w + \epsilon$ from our previous deduction regarding the relationship between s_0 and w. This therefore implies that $l \leq w$, which means that w, which is a lower bound of S is larger than or equal to any other possible lower bound of S. Therefore, by definition of infimum, we know that $w = \inf S$. □

Problem 2

If $0 < a < b$, determine $\lim_{n\to\infty} \frac{a^{n+1}+b^{n+1}}{a^n+b^n}$.

Solution:

$$\begin{aligned}
\lim_{n\to\infty}\left(\frac{a^{n+1}+b^{n+1}}{a^n+b^n}\right) &= \lim_{n\to\infty}\left(\frac{b^n\left(\frac{a^{n+1}}{b^n}+b\right)}{b^n\left(\frac{a^n}{b^n}+1\right)}\right)\\
&= \lim_{n\to\infty}\left(\frac{\frac{a^{n+1}}{b^n}+b}{\frac{a^n}{b^n}+1}\right)\\
&= \lim_{n\to\infty}\left(\frac{a\left(\frac{a}{b}\right)^n+b}{\left(\frac{a}{b}\right)^n+1}\right)\\
&= \frac{a\lim_{n\to\infty}\left(\frac{a}{b}\right)^n+b}{\lim_{n\to\infty}\left(\frac{a}{b}\right)^n+1}
\end{aligned}$$

Now, since $0 < a < b$, $\frac{a}{b} < 1$. Therefore, we know that $\lim_{n\to\infty}\left(\frac{a}{b}\right)^n = 0$. Thus, we can conclude that:

$$\begin{aligned}
\frac{a\lim_{n\to\infty}\left(\frac{a}{b}\right)^n+b}{\lim_{n\to\infty}\left(\frac{a}{b}\right)^n+1} &= \frac{a\cdot 0+b}{0+1}\\
&= \frac{0+b}{1}\\
&= \frac{b}{1}\\
&= b
\end{aligned}$$

Thus, $\lim_{n\to\infty}\left(\frac{a^{n+1}+b^{n+1}}{a^n+b^n}\right) = b$.

Problem 3

Let $x_1 = 1$ and let $x_{n+1} = \sqrt{2 + x_n}, \forall n \in \mathbb{N}$ such that $n \geq 1$. Show that $\lim_{n\to\infty}(x_n)$ exists and find the limit.

Solution:

(I) Prove (x_n) is upperbounded.

Selecting 10 as an arbitrary number I conjecture upperbounds the sequence (x_n), I will attempt to prove that 10 does indeed upperbond (x_n) by induction. Let $P(n)$ be the statement $P(n) : x_n \leq 10$.

(a) Show the base case holds (Show $P(1)$ is true).

LHS:	RHS:
$= x_1$	$= 10$
$= 1$	$= 10$

Therefore, since the LHS $\leq$ RHS, the base case holds and $P(1)$ is true.

(b) Formulate the induction hypothesis (State $P(k)$).

Let $k \in \mathbb{N}$ be an arbitrary natural number for which our statement holds true. We can therefore state our induction hypothesis to be:

$$P(k) : x_k \leq 10$$

(c) Prove the induction step (Show $P(k) \implies P(k+1)$).

$$\begin{aligned}
x_{k+1} &= \sqrt{2 + x_k} && \langle \text{ By definition of } (x_n) \rangle \\
&\leq \sqrt{2 + 10} && \langle \text{ From our induction hypothesis } \rangle \\
&= \sqrt{12} \\
&\leq 10
\end{aligned}$$

Therefore, since $P(1)$ is true, and for arbitrary $k \in \mathbb{N}$ where $P(k)$ holds true, $P(k) \implies P(k+1)$, then by the principle of mathematical induction, $P(n)$ is true $\forall n \in \mathbb{N}$ and thus 10 is an upper bound of (x_n).

(II) Prove that (x_n) is increasing.

By computing the first few values of (x_n), I conjecture, and will attempt to prove by induction, that (x_n) is an increasing sequence. For this purpose, let $P(n)$ be the statement $P(n) : x_n \leq x_{n+1}$

(a) Show the base case holds (Show $P(1)$ is true).

$$\begin{aligned} &\text{LHS:} & \text{RHS:}& \\ &= x_1 & = x_2& \\ &= 1 & = \sqrt{2+1}& \\ &= 1 & = \sqrt{3}& \end{aligned}$$

Therefore, since the LHS $\leq$ RHS, the base case holds and $P(1)$ is true.

(b) Formulate the induction hypothesis (State $P(k)$).

Let $k \in \mathbb{N}$ be an arbitrary natural number for which our statement holds true. We can therefore state our induction hypothesis to be:

$$P(k) : x_k \leq x_{k+1}$$

(c) Prove the induction step (Show $P(k) \implies P(k+1)$).

$$\begin{aligned} x_{k+2} &= \sqrt{2 + x_{k+1}} \qquad \langle \text{ By definition of } (x_k) \rangle \\ &\geq \sqrt{2 + x_k} \qquad \langle \text{ By induction hypothesis } \rangle \\ &= x_k \end{aligned}$$

Therefore, since $P(1)$ is true, and for arbitrary $k \in \mathbb{N}$ where $P(k)$ holds true, $P(k) \implies P(k+1)$, then by the principle of mathematical induction, $P(n)$ is true $\forall n \in \mathbb{N}$ and thus (x_n) is an increasing sequence.

Thus, by **(I)** and **(II)**, (x_n) is an increasing sequence which is bounded above. Therefore, by the Monotone Convergence Theorem, (x_n) converges and thus $\lim_{n\to\infty}(x_n)$ exists.

(III) Find $\lim_{n\to\infty}(x_n)$.

Let $L = \lim_{n\to\infty}(x_n)$. Trivially, we have that $\lim_{n\to\infty}(x_{n+1}) = L = \lim_{n\to\infty}(x_n)$, and thus:

$$
\begin{aligned}
\lim_{n\to\infty} x_{n+1} &= \lim_{n\to\infty} \sqrt{2+x_n} && \langle \text{ By definition of } (x_n) \rangle \\
\lim_{n\to\infty} x_{n+1} &= \sqrt{2+\lim_{n\to\infty} x_n} && \langle \text{ By Algebraic Limit Theorems } \rangle \\
L &= \sqrt{2+L} \\
L^2 &= 2+L \\
L^2 - L - 2 &= 0 \\
(L+1)(L-2) &= 0
\end{aligned}
$$

Thus, the limit of (x_n) must be one of either $L = -1$ or $L = 2$. However, since (x_n) is an increasing sequence, and $x_1 = 1$, it naturally must be that $\lim_{n\to\infty}(x_n) \geq 1$. Thus, $\lim_{n\to\infty}(x_n) = 2$.